amount of heat energy from the Sun, but they react differently. It is because both of them have different properties. In general, the amount of heat energy absorbed or lost by a body is determined by three factors.

1. Mass of the body
2. Change in temperature of the body
3. Nature of the material of the body

We can understand this from the following observations.

Observation:1

Quantity of heat required to raise the temperature of 1 litre of water will be more than the heat required to raise the temperature of 500 ml of water. If Q is the quantity of heat absorbed and m is the mass of the body, then

$Q \alpha m$ (7.1)

Observation: 2

Quantity of heat energy (Q) required to raise the temperature of 250 ml of water to 100°C is more than the heat energy required to raise the temperature to 50°C. Here, $Q \alpha \Delta T$, where ΔT is the change in temperature of the body.

Thus, heat lost or gained by a substance when its temperature changes by ΔT is,

$$Q \alpha m\Delta T$$

$$Q = mC\Delta T \quad (7.2)$$

From the above equations, the absolute temperature and energy of a system are proportional to each other. The proportionality constant is the specific heat capacity (C) of the substance.

$$\therefore C = Q/m\Delta T$$

Thus, specific heat capacity of a substance is defined as the amount of heat required to raise the temperature of 1 kg of the substance by 1^0C or 1 K. The SI unit of specific heat capacity is $Jkg^{-1} K^{-1}$. The most commonly used units of specific heat capacity are J/kg°C and J/g°C.

Among all the substances, water has the highest specific heat capacity and its value is 4200 J/kg°K. So, water absorbs a large amount of heat for unit rise in temperature. Thus, water is used as a coolant in car radiators and factories to keep engines and other machinery parts cool. It is because of this same reason, temperature of water in the lake does not change much during day time.

Problem 3

Calculate the heat energy required to raise the temperature of 2 kg of water from 10°C to 50°C. Specific heat capacity of water is 4200 $JKg^{-1} K^{-1}$.

Solution:

Given m = 2 Kg, $\Delta T = (50^o - 10^o) = 40^oC$
In terms of Kelvin, $\Delta T = (323.15 - 283.15) = 40K$, $C = 4200\ J\ Kg^{-1} K^{-1}$

$\therefore$ Heat energy required, $Q = m \times C \times \Delta T$
$= 2 \times 4200 \times 40 = 3{,}36{,}000$ J



Water in its various form, has different specific heat capacities.

Water (Liquid state) = 4200 $JKg^{-1} K^{-1}$
Ice (Solid state) = 2100 $JKg^{-1} K^{-1}$
Steam (Gaseous state) = 460 $JKg^{-1} K^{-1}$

7.5 Heat capacity or Thermal capacity

Now, you are familiar with specific heat capacity. It is the heat required to raise the temperature of a unit mass of a body by 1°C. But, heat capacity is the heat required to raise the temperature of the entire mass of the body by 1°C. Thus, heat capacity or thermal capacity is defined as the amount of heat energy required to raise the temperature of a body by 1°C. It is denoted by C'.

$$\text{Heat Capacity} = \frac{\text{Quantity of heat required}}{\text{Raise in Temperature}}$$

$$C' = Q/T$$

IX_SCI_EM_Unit-07_PHY.indd 85 12/7/2022 6:13:32 PM